

Post-Detection report for 2024 IAA SETI Committee meeting

16 Oct 2024, Milan & online

Report given remotely by Kathryn Denning. (Associate Professor, Anthropology, York University, IAA SETI Committee member since 2005, SETI Institute Science Advisory Board member). kdenning@yorku.ca

Prefatory note

As in my last update, we do not currently have a formally organized Post-Detection Working Group or Subcommittee within the SETI Committee. This may change in 2025-6.

Meanwhile:

- we have had a small task group working on revising the “Declaration of Principles Concerning the Conduct of the Search for ExtraTerrestrial Intelligence (SETI)” (aka Protocols)
- some members and other researchers from outside the Committee have been very active on post-detection-related topics.

Previous post-detection updates to the Committee are available for reference: [2023](#), [2022](#) and [2021](#). Much of that content is still generally applicable. In 2024, interest is intensifying in “discovery and society” topics for extraterrestrial life and intelligence, on issues ranging from optimal public communication to ethics to legal matters to public opinion. However, funding structures have not kept pace with the interest. It bears repeating – again – that post-detection-related interdisciplinary research is essential, but persistently underfunded.

The Committee has much work ahead concerning Protocols updates and related matters.

Overview of today’s update

1. Post-detection Related Abstracts from IAC 2024
2. Some relevant recent and ongoing activities of SETI Permanent Committee members, including new research groups, projects, talk series, conferences and symposia with post-detection relevance, including: A Sign in Space; IAU/Kavli Symposium; IISL SETI & Law Working Group
3. NASA developments: technosignatures, Communicating Discoveries workshop, etc.
4. SETI Post-Detection Hub at St. Andrews
5. The United Nations, Foresight / Futures, and Space
6. The Protocols: a Draft Revised “Declaration of Principles Concerning the Conduct of the Search for ExtraTerrestrial Intelligence (SETI)” for discussion
 - Specific points to ponder: PD Subcommittee resurrection; Best Practices; etc.
 - Notes on history of prior work on Protocols
 - Abstracts of Protocols Task Group’s papers in IAC Baku 2023 and IAC Milan 2024

1. Post-detection Related Abstracts from IAC 2024

from <https://iafastro.directory/iac/browse/IAC-24/A4/>

As ever, all the papers in the SETI sessions, including the technical ones, are relevant to post-detection, because they explore the forms of detection which are now possible, or may be possible in future. Some IAC 2024 contributions more specifically focused on topics related to public reaction and related policy etc. include the following. (Some talks listed below were withdrawn/not presented, but the abstracts are still of interest.)

- **Daniela De Paulis *et al***, KEYNOTE: "Billingham Cutting-Edge Lecture" - Global outreach and cultural impact of A Sign in Space, an interdisciplinary simulation of a First Contact scenario. [Abstract](#).
 - **Chelsea Haramia**, Causal Impotence and Cosmic Messaging: A Logical Response to the Barn Door Argument. [Abstract](#).
 - **Paolo Musso *et al***, The "Toulouse Reboot": Opening a New Era in the Studies on Interstellar Messages. [Abstract](#).
 - **Andrea Harrington and Charles Stotler**, State Responsibility for First Contact Under International Law. [Abstract](#).
 - **Les Tennen *et al***, The Future of the SETI Post-Detection Protocols: Progress Towards Revisions. [Abstract](#).
 - **Raoul Cardellini**, Assessing the consistency between SETI and metalaw and the collective interest of humankind in the exploration and use of outer space. [Abstract](#).
 - **Julia DeMarines and Chelsea Haramia**, An Active and Community-Based Approach to Inclusive Debates in SETI Ethics. [Abstract](#).
 - **Lorenzo De Piccoli**, Technosignatures. The 400-Year Search for Extraterrestrial Artefacts. [Abstract](#).
 - **Giorgio Gavrighi**, Artificial intelligences (AIs) could potentially serve as the catalyst for the first contact between our society and an alien one. [Abstract](#).
 - **Kate Genevieve *et al***, Plurality in Post Detection Scenarios. [Abstract](#).
 - **Nigar Ismayilzada *et al***, Ethical frontiers and societal echoes: navigating the implications of SETI research. [Abstract](#).
 - **Ali Mammadov**, Discovering Extraterrestrial Life in the Universe: Scientific, Technical, and Interdisciplinary Dimensions of Extraterrestrial Life and SETI's contributions to finding Extraterrestrial Life. [Abstract](#).
 - **Samiksha Raviraja *et al***, The Discovery of Alien Life and How That could Impact Fundamental Human Beliefs, Societal Views and Philosophies. [Abstract](#).
 - **Gabriela Radulescu**, The Relational Legacy of Extraterrestrial Intelligence. [Abstract](#)
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2. Some relevant recent and ongoing activities of SETI Permanent Committee members (n.b. This list is far from complete.)

- **The SETI Institute and Breakthrough Listen** continue their extensive public outreach efforts.
- **Daniela de Paulis and team, [A Sign in Space](#)** This award-winning project, presented to the Committee by de Paulis in 2021 and 2022 as work in progress, proceeded in 2023-2024 with the SETI Institute and continues to be a great success, with thousands of citizen participants engaged. Multiple members of the SETI Committee were involved in different capacities.
- **Breakthrough Discuss 2024.** Given that Breakthrough Listen's international HQ [moved to Oxford](#) in autumn 2023, in 2024 the [Breakthrough Discuss](#) meeting was held in Oxford for the first time. The rapid expansion of technosignatures searches, presented by a variety of speakers including SETI Committee members, as always poses some new challenges for post-detection / discovery & society, although the latter are not usually explored in depth in the conference. [Talk Playlist](#). Previous Discuss conferences back to 2016 are [here](#).
- **PSETI (Penn State University).** The [PSETI seminar series](#) run by Jason Wright and colleagues includes SETI & Society / post-detection topics; recordings are available. PSETI sponsors the [Order of the Octopus](#), a large, thriving community of early career researchers in SETI who held an [Assembly in 2024](#) and have a Discord group.

The now-biannual [PSETI Symposium 2025](#) will be 18 - 21 Aug 2025, online + in person at Penn State University Park: *"This symposium will be a broad look at SETI, including new technosignature search results and upper limits, new search methodologies and instrumentation, theory, and social aspects of the field including the post-detection protocols, message construction, and the historical roots of the field. Interdisciplinarity will be emphasized."* Previous PSETI symposia have seen a significant focus on post-detection with talks and break-out sessions (from Committee members A. Berea, K. Denning, S. J. Dick, J. Tarter, S. Wright, with colleagues C. Haramia, R. Charbonneau, B.S. McConnell, and G. Profitiliotis). We aim to continue that tradition in 2025.
- **[The Technosignatures Seminar Series](#)** at NASA Goddard, continues to feature speakers on a variety of technosignatures topics, including SETI Committee members, and some talks directly relevant to post-detection. The website has a past and future speaker list and recordings of presentations. To join the mailing list, contact Committee member Ravi Kopparapu: ravikumar.kopparapu@nasa.gov
- **Social and Ethical Frontiers in Space 2024**, a joint conference of the European Astrobiology Institute (EAI), and the Society for Social and Conceptual issues in Astrobiology (SSoCIA), Sept 2024, Finland, involved several papers related to technosignatures and others more generally related to extraterrestrial life.
- **[AbSciCon 2024](#)** Providence, RI, USA, hybrid, May 2024 included sessions on technosignatures, exoplanets, and a large three-part session on "Social Sciences, Humanities, and Astrobiology" with diverse papers including some concerning public communication and potential reactions to discoveries of extraterrestrial life. Multiple Committee members gave papers. AbSciCon sessions and plenaries in 2022 and 2024 addressed standards of evidence for biosignatures, and communication of astrobiology results to broad audiences -- entangled subjects now receiving sustained attention from senior astrobiology scientists and administrators.
- **[Communicating Discoveries in the Search for Life in the Universe](#)**, online, Feb-Mar 2024, from the NASA Astrobiology Program. See "NASA Developments" section below for details.

- **[“Breaking News: We Found Extraterrestrial Life!”](#) workshop at the Lorentz Centre, Leiden University**
Continuing on from the Communicating Discoveries workshop was the “Breaking News: We Found Extraterrestrial Life!” week-long workshop at the Lorentz Centre in Leiden in Sept 2024, led by researchers from Leiden University, the NASA Astrobiology Program, and the Earth-Life Science Institute in Tokyo. This work is expected to continue in the years ahead.
- **[IAU/Kavli Symposium](#) (IAUS 387), “(Toward) Discovery of Life Beyond Earth and its Impact”, 15-19 April 2024, Durham UK, in-person and virtual.** This symposium’s in-depth five-day program included speakers from many fields, including astrobiology, astronomy, law and policy, media, social sciences, philosophy, theology, and futures studies. Multiple members of the IAA SETI Committee (including C. Oliver, J. Haqq-Misra, M. Dominik, L. Tennen, J. Elliott, J. Rummel, D. de Paulis, K. Denning, S. Croft, S. Webb) were co-organisers of the Symposium, presenters, or both. The Symposium proceedings will be published by Cambridge University Press in 2025 and will be of substantial relevance.
- **IISL Working Group on SETI & Law.** In Jan 2024, the International Institute of Space Law established a Working Group on SETI & Law under convenor Michael Bohlander. As of Sept 2024, it will go forward under the leadership of Andrea Harrington and is scheduled to run until 2026. The Working Group intends to “...*contribute to laying the legal groundwork for providing guidance to relevant domestic and international stakeholders in government and civil society, regarding the analysis and development of regulatory protocols.*” Details of the WG’s focus and intended outputs are evolving, but generally, the WG is focused upon legal issues that would follow a confirmed detection of extraterrestrial intelligence. Several IAA SETI Committee members are involved in the Working Group, including M. Garrett, L.I. Tennen, J. Elliott, J. Wright, and K. Denning.

3. NASA Developments

SETI was part of the NASA research portfolio for a few years starting around 4 decades ago, under the second Chair of the SETI Committee, John Billingham, who was also director of Life Sciences at NASA Ames.¹ Infamously, this changed in the early 90s. Now, after a long period of SETI having an ostensibly separate but parallel path with astrobiology in the USA, change is in the air. American researchers have been working on the gradual reintegration of technosignatures research into NASA’s search for extraterrestrial life, starting with the NASA Technosignatures Workshop in 2018. Sessions at the Penn State SETI Symposia in 2022 and 2023 specifically addressed the potential for future NASA missions to include technosignatures research. In May 2024, David Grinspoon, in his first year as new NASA Senior Scientist for Astrobiology Strategy, provided his first Town Hall briefing for the community at [AbSciCon](#), and floated the reintegration of technosignatures into NASA work under a “Future of Life” cross-cutting interdivisional theme. Now, 6 months later, an upcoming Oct 2024 meeting hosted by the NASA Astrobiology Program, *Astrobiology and the Future of Life*, boasts a [program](#) including multiple papers on technosignatures science, and papers on cultural dimensions and social research related to astrobiology and SETI, including some from SETI Committee members.

Also relevant here is the expansion of the NASA Astrobiology Program’s focus to include more structured and intensive thought about communicating potential future discoveries of life beyond Earth to wide audiences. As

¹ Steven J. Dick oral history interviews with John Billingham, 1990-1993.

noted in the [2023](#) and [2022](#) PD updates, NASA Astrobiology has been driving forward for several years with discussions and workshops and [reviews](#) about [confidence in life detection](#) and [Standards of Evidence for biosignatures](#). In 2024 this attention was channeled into a large interdisciplinary virtual workshop, [Communicating Discoveries in the Search for Life in the Universe](#), which involved several members of the Committee and the technosignatures research community, and multiple science journalists who cover SETI.

The collectively-authored CDSLU workshop report is expected to appear in *Astrobiology* in 2025. Meanwhile, as a workshop participant and also as a co-organizer involved in early planning, I offer these observations: The 80+ workshop attendees were selected by organizers from a large pool of applicants, for a balance in terms of professional expertises, career stages, and gender. In addition to astrobiology (and SETI) scientists, there were science communication *researchers*, social scientists, and science communicators (print and new media). Many had been involved in prior instances of putative discoveries of extraterrestrial life, and so brought a wealth of experience to the conversation. Pre-reading materials included: overviews of science communication research (L. Billings), social sciences and astrobiology (K. Denning), and the history of astrobiology and lessons learned (S. J. Dick); brief overviews of astrobiology case studies including GFAJ-1 and ALH84001, and cases from other domains of science (climate science, LIGO); and potential future ETL discovery types (Mars Sample Return; icy moons; exoplanet atmosphere).

Although the meeting was generally oriented towards biosignatures, issues of communicating about the discovery of ETI and ETL have significant overlaps: the challenges are very similar, particularly for the scientists in the public eye in a high stakes ETL/I situation. This meeting generated concrete recommendations which we should consider: e.g. distinguishing between the responsibilities of scientists and those of professional science communicators; and emphasizing the responsibilities of institutions running ETL/I work for supporting their scientists in communication challenges. This community of astrobiology scientists and science communicators and social researchers, balanced by gender and career stage, emphasized that scientists need more support from their institutions and funders; said that they are concerned about their safety in these situations²; noted that more should be learned from prior putative detection situations; and emphasized the need to integrate social research into this endeavour. Full details to follow in the 2025 workshop report.

A connected, very recent development is the NASA Science Mission Directorate [Request for Information of 8 Oct 2024](#), as an initial stage in developing the 2025 NASA Decadal Astrobiology Research and Exploration Strategy. In addition to inviting input on a suite of science topics, the SMD is asking for feedback on:

“Topic 8: Prepare for the Discovery of Life Beyond Earth and Subsequent Post-Discovery Activities. Recognize that finding evidence of life beyond Earth, whether extant or extinct, represents a pivotal evolution in the field of astrobiology. In tandem with community efforts focused on refining the science and mission requirements for life detection, e.g., the Search for Life Mission Analysis Group, SFL-SAG, should be the development of frameworks for international cooperation and public engagement to responsibly handle such discoveries. Build on previous efforts from recent workshops e.g., “Communicating Discoveries in the Search for Life in the Universe” and prepare for activities – including post-discovery science – that consider and respond to the profound implications of life’s existence elsewhere, ensuring the field evolves to address new scientific, ethical, and societal changes.”

Presumably there will be more to report next year along these lines.

² My invited “Social Sciences and Astrobiology: Some Orientation and Future Directions” overview for this meeting briefly covers the issue of safety and provides references / resources. [Available here](#).

4. The SETI Post-Detection Hub at the University of St. Andrews

Coordinator: IAA SETI Committee long-time member John Elliott (john.elliott@st-andrews.ac.uk).

The Hub, a UKSRN initiative, launched in Nov 2022 and is jointly hosted by the St Andrews Centre for Exoplanet Science (led by Committee member Martin Dominik and Emily Finer), the Centre for Global Law and Governance, the Durham Law School, and the Jodrell Bank Centre for Astrophysics of the University of Manchester.

The Hub now includes over 40 researchers from myriad fields – about half in the UK and half international. Some have been working on SETI topics for years and decades, and others are newer to SETI/ technosignatures but have significant relevant expertise e.g. in science communication research, public policy, risk, law, etc. 12 Hub members are also members of the IAA SETI Committee.

In the past year, the SETI Post-Detection Hub has been building on its foundations to drive on with many new initiatives. Funding applications (both project-focused and for supporting the Hub as a research centre) are in process. In addition to research activities, student networks have recently been established (at St Andrews and Internationally) and student projects at all levels of degree: BSc, MSc, and PhD. Hub members participated in the British and Irish Science Festivals, and many were involved in the IAU/Kavli symposium in Durham in April. UK Government interest and discussions also continue – prompted by a question in the UK Houses of Parliament (January 2024) on post detection that followed an interview (Elliott, Garrett from the Hub) in the government’s magazine – The House. Media interest continues to be high, with filming, interviews, and articles.

Central to this year’s endeavours has been work on scenarios – a tool from Futures Studies / strategic foresight for thinking about different possible social and technological futures 5, 10, or 25 years from now. The aim is to contribute to the promotion of *futures literacy* across the broader SETI ecosystem by reframing the implicit assumption that a future detection will take place in a world that is just an extension of today. Led by Futures Studies specialist G. Profitiliotis, the Hub scenarios working group has systematically developed several different scenarios for the future contextual environment of the SETI ecosystem in 2050, capturing relevant potential economic, political, legal, cultural, societal, environmental, scientific, and technological developments.

There are many more PD Hub research and development project initiatives planned and already beginning (including work with the IISL -- an initiative by Elliott & M. Bohlander), to explore ideas and populate resources for post-detection preparedness that cannot be detailed here, given the length limitations.

5. The United Nations, Foresight / Futures, and Space

(Thanks to Dr. George Profitiliotis, United Nations Development Programme Futures Fellow in 2024 – for info.)

Our first C/SETI Committee Chair, astronomer Rudolf Pešek, emphasized at the outset in the 1970s that among the Committee’s top priorities were these: “To interest the United Nations in CETI and help them in all CETI problems”; and “To endeavor – by all means we may have – to establish the rule and make it a practice that all really important decisions concerning CETI are to be taken by the United Nations.”³ The past year brought some

³ R. Pešek 1978. “Activities of the IAA CETI committee from 1965-1976 and CETI outlook” in *Acta Astronautica* vol 6: 3-9. Pergamon Press. [Link](#).

changes on the UN level which may intersect with the search for ETL/ETI, and which resonate with the IAA SETI Committee's history and original intentions.

Specific recent UN developments of interest include these:

- The United Nations Development Programme ([UNDP](#)) Strategy and Futures Team of the Executive Office of the Administrator
 - published (in 4 languages) its [Signals Spotlight 2024](#) annual report which explicitly mentions that: *"if and when extraterrestrial life were discovered, it would require new forms of inter-species interaction and communication beyond what we know today. This will pose ethical challenges as well as technical ones"* (p. 33). It also included a dedicated outer space theme (p. 62).
 - published its [2024 Trends Report](#), which for the first time included trends on outer space, placing space squarely within the broader development landscape (p. 26).

As a UNDP Futures Fellow, Profitiliotis contributed subject-matter insights to both reports.

- During the [UN Summit of the Future in Sept 2024](#), world leaders adopted a [Pact for the Future](#), which, before adoption, underwent 4 revisions via institutional consultation avenues.
 - Profitiliotis contributed language change suggestions related to "exploration" and "the search for extraterrestrial life" during these consultations.
 - Explicit mention of "exploration" was accepted in the adopted text which contains a dedicated Action (#56) on outer space: *"We will strengthen international cooperation for the exploration and use of outer space for peaceful purposes and for the benefit of all humanity"* (p. 39). The same action "encourages the Committee on the Peaceful Uses of Outer Space to further consult on the proposal to hold a fourth [United Nations Conference on the Peaceful Exploration of Outer Space](#) (UNISPACE IV) in 2027". ([Proposal here](#).)
 - The adopted Pact for the Future also includes a [Declaration on Future Generations](#) with a commitment to act on *"Fostering an organizational culture that is future-oriented and mainstreamed across the United Nations system in order to facilitate science-and evidence-based decision-making by developing diverse capabilities, including anticipatory planning, foresight and futures literacy"* (p. 60). This is already underway in the recently created [UN Futures Lab Network](#).
- The UN Secretary-General recently formed a [Scientific Advisory Board](#) to provide insights on governing the benefits and risks of breakthroughs in science and technology, with an initial focus on artificial intelligence, biotechnology, and climate change. The SAB released a Sept 2024 statement on [Trust in Science](#) (worth reading!) which underscores: the importance of "horizon scans of developments in science and technology" (a strategic foresight method); the need for "public repositories of scientific research that are easily accessible, transparent, and understandable in plain-language"; and the need for "international scientific panels on AI and other potentially disruptive technologies for regular, reliable, and unbiased scientific assessments that inform policies and governance." One can imagine potential future links to searches for ETL/ETI.

The SETI Committee should watch these developments, and engage, in the years ahead.

6. The Protocols: a Draft Revised “Declaration of Principles Concerning the Conduct of the Search for ExtraTerrestrial Intelligence (SETI)”

As reported in 2023: from late 2022 onwards, a small task group (Carol Oliver, Kathryn Denning, Michael Garrett, Les Tennen) started work on the preliminaries for a revision of the Protocols, building on work done by SETI Committee members in recent years. The 2023 updates on this work, and associated files, are [here](#). The abstract is included at the end of this document.

Work has continued, and SETI Committee members should by now have received a copy of the 1 October 2024 Draft Revised “Declaration of Principles Concerning the Conduct of the Search for ExtraTerrestrial Intelligence (SETI)”. The authors of this Draft are IAA SETI Committee members Leslie I. Tennen, Kathryn Denning, Carol Oliver, Michael A. Garrett. n.b. This Draft is intended for discussion and not every author has given blanket endorsement to all the wording. Further revisions will reflect SETI Committee input.

This Draft will be presented by L. Tennen at the Milan IAC 2024 in the [SETI & Society](#) session on 15 Oct 2024. There will also be a brief presentation by Tennen and initial discussion during the IAA SETI Committee meeting on 16 Oct 2024 (hybrid: online and at IAC).

The current 1 Oct 2024 Draft has significant updates which will be covered by L. Tennen in his presentation.

We want your feedback. Additional supporting materials and details about the feedback process will be sent to Committee members in the weeks ahead. The team – the IAA SETI Executive (M. Garrett, A. Siemion, C. Oliver, L. Walton) with L. Tennen and K. Denning – will review your input. Then the Draft will be further revised, recirculated, and discussed. Ultimately a final Draft will be brought to a Committee vote in mid-2025.

Speaking individually after several years thinking about Protocols updates: personally, I think focus on two key points for initial discussion may be useful as the Committee collectively considers revisions/next steps:

a) Potential Best Practices Guidelines which would accompany the Declaration of Principles. The preamble of the 1 Oct 2024 Draft says *“this Declaration will be supplemented with Best Practices Guidelines and a Code of Conduct that will be periodically re-examined and updated”*. The idea for Best Practices – first floated by a group of IAA SETI Committee authors in [2019](#) – is that these could be less formal than the Declaration of Principles, readily produced and updated by small working groups. These guidelines could, for example, include Best Practices for projects with students, BPs for projects with citizen scientists, BPs for data handling, BPs for communicating about technosignatures to broad audiences. Committee members will have additional ideas worth considering.

b) Post-Detection Sub-Committee. Principle 7 of the 1 Oct 2024 Draft states: *“The IAA SETI Committee will appoint a Post-Detection Sub-Committee to assist in matters that may arise in the event of a confirmed signal, and to support the scientific and public analysis by offering guidance, interpretation, and discussion of the wider implications of the detection.”* This Principle would commit us to re-establishing a PD Sub-Committee after a 10-year hiatus. PD Sub-Committees existed from about 1996 to 2014, with varying levels of activity and remits.

In considering a new PD Sub-Committee and its potential remit, I believe we should carefully consider current overall Committee composition. The Committee has many technical experts and astronomers. However, currently, of ~95 members: few are from countries other than USA + UK/Europe; only ~16 are female; almost none are ECRs; only two are legal experts; ~11 are in humanities/social science; we have few media professionals or professional science communicators. We do not presently have a critical mass of “SETI & Society” experts within the Committee. An intentional approach to membership renewal would be needed to build capacity as well as diversity/equity.

Additionally, in considering a new PD Sub-Committee and its potential remit, I believe we should also carefully consider the limitations and actual capacities of a subcommittee with no funding of its own. This has often been raised in the past. Over 25 years ago, the Terms of Reference for the SETI Post-Detection Science & Technology Subcommittee specified that should the interdisciplinary Subcommittee be called upon to assist in detailed analysis or advice concerning a putative signal of extraterrestrial origin, “All costs for such activities of the Subcommittee would be born by the organization making the detection.”⁴ That provision emerged in an era when SETI searches were few, false alarms were not frequent, and when the Subcommittee would not be expected to help manage a fast-paced social media news environment.

Similarly, one of the original architects of the 1989 Declaration of Principles (from which the idea of a post-detection sub-committee originally emerged), [Allan E. Goodman](#), realized 35 years ago that Post-Detection Protocol-related matters required both some support from the scientific organizations conducting SETI experiments, and, simultaneously, some independence from them. The elegant solution he proposed was the creation of a standing consultative commission composed of members of governments and private organizations conducting SETI, scientists and interdisciplinary researchers, who would serve five-year terms without compensation, but with support in the forms of funding for travel to meetings, and administrative support for organizing meetings and preparing commission reports. Goodman suggested that “These administrative costs should be financed by an endowment raised from donations by governments and private persons and foundations.”⁵ Today, when SETI policy questions are ever more complex, this remains a valuable suggestion.

History of Prior Work on SETI Post-Detection Protocols

The history of the Protocols is under-appreciated. Although some current members of the SETI Committee have been involved since the first version of the Protocols 35-40 years ago, many current members were not present for either the development of the 1989 version or the 2010 revision. Anyone who would like to learn more / access resources is welcome to email kdenning@yorku.ca.

This history is briefly reviewed in the Task Group’s papers Oliver et al 2023 and in Tennen et al 2024. Here is a **sketch of the history for easy reference**, linking to documents available on the [IAASETI home page](#):

- Work in the 1980s culminated in the 1989 “[Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence](#)” (also: [cover letter](#) + [Declaration](#)), approved by the IAA SETI

⁴ Terms of Reference, “SETI Post-Detection Science & Technology Subcommittee of the IAA SETI Committee.” 2000. [Link](#).

⁵ Allan E. Goodman 1990. “Diplomatic and Political Problems Affecting the Formulation and Implementation of an International Protocol for Activities Following The Detection of a Signal from Extraterrestrial Intelligence.” *Acta Astronautica* Vol. 21, No. 2, pp. 103-108.

Committee, and by 1992 also endorsed by IAA Board of Trustees, IISL, IAF, COSPAR, IAU, and Commission J of URSI.

- In 1995, a companion document was produced: the “[Draft Declaration of Principles Concerning Sending Communications with Extraterrestrial Intelligence](#)”. In 1996 a Position Paper, “[A Decision Process for Examining the Possibility of Sending Communications to Extraterrestrial Civilizations](#)” attached both the 1989 “Declaration of Principles Concerning Activities Following the Detection...” and the 1995 “Draft Declaration of Principles Concerning Sending Communications...”: the Position Paper was approved by the IAA BoT and the IISL. In 2000 the Position Paper was presented to UN COPUOS and noted in the COPUOS session report approved by the UN General Assembly. A [2007 revision of the 1996 Position Paper](#) within the SETI Committee contained updates.
- In 2010, the revised “[Declaration of Principles Concerning the Conduct of the Search for Extraterrestrial Intelligence](#)” updated / superseded the 1989 Declaration, and also elided the specific 1989 Principle #8 reference to a “separate agreement, declaration or arrangement” concerning consultation procedures for sending responses. The 2010 Revised Declaration was approved by the IAA SETI Committee.

This history is important for orientation as we consider revisions, and for appreciation for the foundational work done by Committee member Michael Michaud, former Chairs John Billingham, Jill Tarter, Seth Shostak, and others. But it’s important for additional reasons, too, as I explain below.

Work on the Protocols revisions (now and in the future), and related work (e.g. IISL SETI & the Law group), would be *greatly* facilitated by better history resources that compile and distill previous work at the SETI & society / policy intersection over the past 50+ years. These resources are currently scattered and, in some cases, perilously close to simply being lost. Some Protocols-related papers were published in IAC proceedings and in *Acta Astronautica*, some early papers are also on the NTRS, and some SETI Committee minutes are available online. However, Committee reports are not all available in published form, SETI Committee archives are not easily accessible, and generally, workshop reports are hard to find. (One example is *Social Implications of the Detection of an Extraterrestrial Civilization* (1999) – the report of a series of three workshops held in Santa Cruz in 1991-2 involving multiple Committee members, edited by former SETI Committee Chair John Billingham and colleagues. These workshops were closely linked to the formation of the Post-Detection Subcommittee in the ‘90s. The report is hard to find.) Our member Steven J. Dick, former NASA Chief Historian, conducted oral history interviews with the SETI community with funding from the SETI Institute in 1992-93; these, too, are little-known resources but contain important contextual information about the original Protocols and their developers.

SETI Committee Secretary Lori Walton continues to work hard as an independent volunteer on SETI Committee history, and to produce valuable papers. But overall, there is an urgent and serious need for better support for historical work in this area: urgent both because the most senior members of our community are retiring and passing away, and because better international protocols concerning the discovery of life beyond Earth (intelligent or otherwise) are very much ‘in the air’ currently (initiatives beyond the SETI Committee, i.e. within NASA, within IISL, etc). Researchers interested in compiling archives, collecting interviews, and building history resources, include: Committee members Kathryn Denning, Lori Walton, Jason Wright, Steven J. Dick; and Rebecca Charbonneau (Historian, American Institute of Physics, and PD Hub), and Kate Genevieve (PD Hub). We would welcome allies and support.

For the record, the abstracts of the Protocols Task Group’s 2023 and 2024 papers are included below.

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The Future of the SETI Post-Detection Protocols: Progress Towards Revisions

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Abstract

The International Academy of Astronautics has been the leading source of guiding principles to be followed in the event of a putative SETI signal detection. After many years of work, in 1989, the IAA formulated a set of Protocols, the “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence”, which received widespread acceptance by the international scientific community. The IAA prepared a supplemental set of draft protocols in 1995 relating to the possible reply to a confirmed communication from an ETI, and both documents were presented in a position paper to the UN Committee on the Peaceful Uses of Outer Space.

The Search for Extra-Terrestrial Intelligence takes place in a dynamic context, and the Protocols are always a “work in progress” that must adapt to and reflect constantly changing scientific and social environments. Thus, the IAA SETI Protocols were streamlined and updated in 2010 after a multi-year process. Subsequently, the IAA SETI Committee has sought to review and revise the Protocols to keep pace with the expansion of search methods, the increased number of facilities and personnel conducting search activities worldwide, and the rapidly evolving information environment. In 2022, a Task Group was formed for this purpose, and a preliminary report was presented the following year during the 74th International Astronautical Congress in Baku. The relevance of this endeavour has been underscored in the past year by the deepening interest in what happens after the discovery of life beyond Earth, as evidenced by the 2024 workshops and symposia sponsored by the NASA Astrobiology Program, and the IAU/Kavli Foundation, the work of the SETI Post-Detection Hub at the University of St. Andrews, and the working group on “SETI and the Law” newly established by the International Institute of Space Law.

This report updates the activities of the Task Group following the 2023 Baku IAC, including further community consultations and proposed text, as we work towards a combination of timeless Principles reflecting the original intentions of the 1989 Protocols and supplementary Codes of Conduct and Best Practices documents that can provide additional guidelines that can be updated regularly to respond to changing circumstances.

Revising the SETI Post-Detection Protocols for the 2020s and Beyond: A Report on Work in Progress.

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Abstract: In 1989, the International Academy of Astronautics (IAA) adopted the “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence”, also known as the Search for Extraterrestrial Intelligence (SETI) Post-Detection Protocols. These provided a set of guiding principles to be followed in the event of a putative signal detection, having been developed by the IAA SETI Permanent Committee over a period of several years starting in the late 1980s, and subsequently approved by the International Institute of Space Law (IISL), and endorsed by the Committee on Space Research (COSPAR), the International Astronomical Union (IAU), and the (International Astronautical Federation (IAF). A streamlined update to reflect the emergence of the Internet was developed and accepted by the IAA SETI Permanent Committee in 2010 as the “Declaration of Principles Concerning the Conduct of the Search for Extraterrestrial Intelligence”.

Since then, the nature of global communication has been dramatically changed by social media, and multiple SETI ‘false alarms’ have demonstrated the ways in which information moves in our present era. Simultaneously, SETI searches – now also known techno-signature searches – have radically expanded in extent, scope, and power, with substantial search efforts underway at Breakthrough Listen, the SETI Institute, the Chinese Five-hundred-meter Aperture Spherical Telescope (FAST), and many other observatories around the world, between them using many of the world’s most powerful telescopes. Moreover, upcoming next-generation telescopes will allow the detection of molecules indicative of the existence of life in exoplanet atmospheres, conduct high cadence all-sky surveys and be able to detect the equivalent of an airport radar 100 light years away. As a result, the community of scientists engaged in SETI research is expanding significantly. In addition, other methods of detecting exoplanets as potential techno-signature targets are increasing, broadening the community of scientists who may not regard themselves as SETI-related but could potentially uncover data of SETI interest.

In 2022, the IAA SETI Committee decided that a high priority was to update the 2010 IAA Declaration of Principles. A proposal for a small task group with the sole task of updating these Principles was initiated at the end of 2022, and co-chairs put in place to develop that idea. Given rapid changes in SETI/technosignatures research and the dynamic communications landscape, we now envision an eventual combination of timeless Principles which generally preserve the original intentions of the 1989 Protocols and Codes of Conduct and Best Practices documents to provide additional regularly updated guidelines, responsive to changing circumstances. We present a report on the progress on the formation of the task group and its deliberations.